

Network Metrics Problem Set

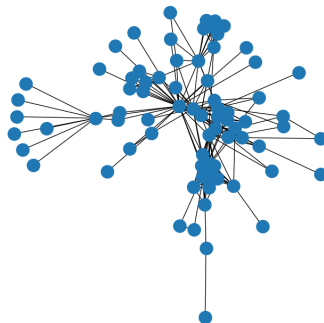


Figure 1: Les Misérables network.

You can also find interesting networks at **The Network Repository** ([link](#)) . Another well known small network is the Les Misérables network, which records character interactions in the famous novel. If you get the data from the link above, note the first line is a comment and the second line records the total number of nodes, 77, and links 254 so should be skipped. The third column records the edge weight.

Problem 1: Download the Les Misérables data from the KONECT database and colour the nodes according to degree, eigenvector, closeness and betweenness centrality.

Problem 2: Since this is a weighted network we can compute the centrality using A_{ij} or W_{ij} . Compare the weighted and unweighted centrality scores.

Problem 3:

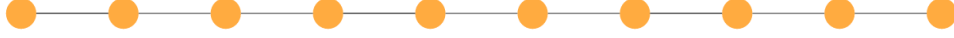


Figure 2: Line graph

Consider a line graph, as shown above, with N nodes.

- (i) What is the betweenness centrality of some node i ?
- (ii) What value of i maximises betweenness?

Hint: split the graph into the part before node i and the part after.

Problem 4: Let

$$n_{ij} = \sum_k A_{ik}A_{kj}$$

This is the **number of common neighbours** of i and j .

(i) Show this is the dot product of row i with row j of the adjacency matrix, where $\vec{A}_i$ denotes the i^{th} row considered as a vector.

(ii) The cosine similarity $\cos \theta_{ij}$ is defined through

$$\vec{A}_i \cdot \vec{A}_j = |\vec{A}_i||\vec{A}_j| \cos \theta_{ij} \quad (1)$$

Write an expression for $\cos \theta_{ij}$ in terms of n_{ij} , k_i and k_j .

Hint: $(A_{ij})^2 = A_{ij}$ since the values are 0 or 1.

(iii) Write an expression for the distance

$$|\vec{A}_i - \vec{A}_j|^2$$

in terms of n_{ij} , k_i and k_j .

Problem 5: The formula for assortativity given in the lectures is quite confusing, we will work out a different, and in my opinion, clearer formulation in this exercise. This is quite long, so feel free to skip it!

(i) Let q_k be the probability a node on the end of a link has degree k . We can count this using a delta function $\delta(k, k_i)$ which is 1 only when the node degree k_i equals the target degree k . The term $A_{ij}\delta(k, k_j)$ is non-zero only when there is an edge from i to j and the degree of j is k . Summing over all edges (twice)

$$N_k = \sum_{ij} A_{ij} \delta(k, k_j) \quad (2)$$

gives the number of edges ending in a degree k node. Find C such that

$$q_k = CN_k \quad \& \quad \sum_k q_k = 1 \quad (3)$$

(ii) Show

$$\sum_k k q_k = \frac{1}{2L} \sum_j k_j k_j$$

(iii) Let $e_{kk'}$ be the probability for a link to have a degree k node on one end and degree k' on the other. The number of links joining a degree k to a degree k' is

$$N_{kk'} = \sum_{ij} A_{ij} \delta(k_i, k) \delta(k_j, k')$$

where the term in the summation is non-zero only if i is connected to k , the degree of i is k_i and of j is k_j . Dividing by the total number of links gives $e_{kk'} = \frac{N_{kk'}}{2L}$. Show

$$\sum_{kk'} k e_{kk'} k' = \frac{1}{2L} \sum_{ij} k_i A_{ij} k_j \quad (4)$$

(iv) Consider a value x_i associated to each node. The average value of x at the end of a node is

$$\mu_x = \frac{1}{2L} \sum_{ij} A_{ij} x_i = \frac{1}{2L} \sum_i k_i x_i$$

The covariance of x_i with x_j over the edges is

$$\text{cov}(x_i, x_j) = \frac{1}{2L} \sum_{ij} A_{ij} (x_i - \mu_x)(x_j - \mu_x)$$

Show

$$\text{cov}(x_i, x_j) = \frac{1}{2L} \sum_{ij} \left(A_{ij} - \frac{k_i k_j}{2L} \right) x_i x_j \quad (5)$$

(v) Using all the results found, show

$$\sum_{kk'} k k' (e_{kk'} - q_k q_{k'}) = \frac{1}{2L} \sum_{ij} k_i \left(A_{ij} - \frac{k_i k_j}{2L} \right) k_j \quad (6)$$

This is the numerator of the degree assortativity. It is a standard result in statistics that normalizing the covariance by the variance gives the **Pearson correlation**. Thus the assortativity coefficient just measures the correlation between the degree at one end of a link with the degree at the other end.